

END TERM EXAMINATION

SIXTH SEMESTER [BCA] MAY-JUNE, 2025

Paper Code: BCA-304

Subject: E-Commerce (Batch 2021 Onwards)

Time: 3 Hours

Maximum Marks: 75

Note: Attempt all questions as directed. Internal choice is indicated.

Q1 Attempt **any five** of the following questions:

(5x5=25)

- (a) Differentiate between e commerce & e business.
- (b) Explain advantages of social commerce.
- (c) Describe different tools for SEO.
- (d) What is the significance of recommendation systems in ecommerce sites, explain.
- (e) Distinguish between credit card and debit card.
- (f) Discuss some demerits of cryptocurrencies.
- (g) Discuss some frequently committed cybercrimes.
- (h) Explain Virtual Private Network(VPN).

Q2 (a) How does a traditional commerce different from ecommerce. State some advantages of ecommerce. (6.5)

(b) Discuss the various unique features of ecommerce. (6)

OR

Q3 (a) "Modern market is driven by ecommerce" -give your views to approve/disapprove the quoted statement. (6.5)

(b) Explain different technological building blocks of ecommerce applications. (6)

Q4 (a) Define SEO. Explain the importance of SEO in the context of ecommerce. (6)

(b) What is a product description page? Explain different parts of the page in brief. (6.5)

OR

Q5 What is third party integration? Explain the concept of data layer integrations for analytics in ecommerce giving suitable examples. (12.5)

Q6 (a) Write brief notes on:

(i) Payment gateway (ii) Peer to peer payment systems

(b) Explain different digital token-based payment systems. (6.5)

OR

Q7 (a) Explain the working of electronic fund transfer in an ecommerce environment. (6.5)

(b) Discuss some major financial instruments. (6)

Q8 (a) Elucidate on different credit card frauds happened in recent times. (5.5)

(b) Elaborate on the major security concerns on social network platforms in connection with cybercrimes. (7)

OR

Q9 (a) Define digital certificates and digital signatures. Write a note on different types of encryption. (6)

(b) What do you understand by securing a channel? Explain some channel securing Techniques. (6.5)

END TERM EXAMINATION

SIXTH SEMESTER [BCA] MAY-JUNE, 2025

Super Code: BCA-306

Time: 3 Hours

Note: Attempt all questions as directed. Internal choice is indicated.

Subject: Internet of Things

Maximum Marks: 75

1. Attempt any Five of the following questions:

(5x5=25)

- What are the sources of IoT?
- Explain the technology behind IoT.
- Differentiate between CoAP and MQTT.
- What are sensors? Classify the sensors based on input, output and data types.
- Define Actuators. Specify the types of actuators with their advantages and disadvantages.
- Explain the architecture of basic embedded system.
- What are the different sensor deployment strategies? Explain using examples.
- List the components and communication technologies required to automate smart cards using IoT.

- Define Internet of Things. What are the visions and characteristics of IoT? (5)
- With the help of neat diagrams, describe the levels of IoT with an example each. (5)
- Differentiate between M2M and IoT. (2.5)

OR

- Diagrammatically explain the architectural view of IoT. Also explain the stages or layers of IoT architecture. (6.5)
- Explain the wired and wireless communication technologies used in IoT. (6)

- What is Wireless Sensing Networks (WSN). Explain the components and each layer of WSN architecture. (6.5)
- Write short notes on (Any Two): (6)
 - Arduino uno
 - Raspberry Pi
 - Beagle Bone

OR

- Explain the principle of RFID using neatly labelled diagram. Also specify the advantages and disadvantages of RFID in IoT applications. (6.5)

- Differentiate between (Any Two): (6)
 - Embedded Systems and IoT
 - Microcontroller and Microprocessor
 - Sensor and Smart sensor

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- Identify and explain the issues while communication and accessing data in wireless medium through IoT devices. (6)
- List the IoT devices required for blinking of LED using Arduino Uno. Explain using circuit diagram. Also specify the functions required to compile and run the code successfully (6.5)

OR

- Explain in detail the survey routing protocols used by routers to communicate in wireless sensor network. (6)
- What is a sketch and its extension? Explain the components and working of Arduino IDE using neatly labelled diagram? (6.5)

- Is IoT prone to developmental and designing challenges? If yes, justify your answer using appropriate statements. (6.5)
- Design the circuit diagram for smart switch in home automation. Explain each device to be used, communication technologies and network protocols applicable for the same. (6)

OR

- What are the security challenges in the world of IoT? How can they be resolved? (6.5)
- Design the circuit diagram for automating street lights in smart city. Explain each device to be used, communication technologies and network protocols applicable for the same. (6)

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SIXTH SEMESTER [BCA] MAY-JUNE, 2025

Paper Code: BCA-304

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Time: 3 Hours

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SIXTH SEMESTER [BCA] MAY-JUNE, 2025

Paper Code: BCA-302

Subject: Data Ware Housing & Mining

Time: 3 Hours

(Batch 2021 onwards)

Maximum Marks: 75

Note: Attempt any five questions in all including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt any Five of the following questions:

(5x5=25)

- (a) What is data mart? Explain types of data mart.
- (a) Briefly explain the ETL process.
- (b) List different components of DW.
- (c) Explain information package diagram(IPD) with an example.
- (d) Differentiate between MOLAP and ROLAP.
- (e) Briefly explain what is understood by Data Mining and KDD?
- (f) What do you mean by data transformation and data discretization?
- (g) What is a Recommender System?

Unit-I

Q2 Explain three tier Architecture of DataWare house and How this architecture helps in better analysis of Business. (12.5)

OR

Q3 Why do big organizations and companies need a data warehouse system? Explain its compelling need and also write differences in a database system and a data warehouse system. (12.5)

Unit-II

Q4 What is Multidimensional modeling in DW and Explain star , snowflake and fact constellation Schema for multidimensional data model with their advantages and disadvantages. (12.5)

OR

Q5 What are the fundamental differences between OLAP and OLTP? What are the key operations involved in OLAP, and how do these operations facilitate efficient data analysis and decision-making in business intelligence systems? (12.5)

Unit-III

Q6 (a) Explain all Phases used for Data Preprocessing. (7.5)
(b) List Various technologies used for Data Mining and what kind of pattern can be mined. (5)

OR

Q7 Discuss Apriori Algorithm and Association Rule Mining. What are the key steps involved Min the Apriori algorithm for association rule mining, discuss with an example and how does it contribute to efficient market basket analysis? (12.5)

Unit-IV

Q8 Explain the significant role of data mining in the retail and telecommunication industries, supporting your explanation with examples from each sector. (12.5)

OR

Q9 (a) Explain the difference between classification and clustering technique, Write steps to classify the IRIS dataset. (7.5)
(b) Explain any two open source data mining tools. (5)

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SIXTH SEMESTER [BCA] MAY-JUNE, 2025

Paper Code: BCAT-312

Subject: Data Visualization & Analytics

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five Questions including Q.No1 which is compulsory.
Select one question from each unit.

Q1. Answer any Ten of the following.

(2.5×10=25)

- i. What is the difference between sample and population?
- ii. Name different sources for data collection.
- iii. Name different types of data elements.
- iv. What do you understand by imputation?
- v. Name few methods for outlier detection.
- vi. Differentiate between parametric and non-parametric hypothesis testing.
- vii. What is Population Distribution difference between Sampling Distribution.
- viii. What is the difference between label encoding and one hot encoding?
- ix. What do you understand by stratified random sampling?
- x. Name various data standardization methods.
- xi. Name the categories for Seaborn visualization function.
- xii. How would you modify the appearance of a widget in Tkinter using style classes?

UNIT-I

Q2.

- i. Explain analytics process model in detail with diagram. (6.5)
- ii. Discuss different profiles where analytics have been used. (6)

OR

Q3.

- i. When measuring the height of all students at a local university it was found that it was normally distributed with a mean height of 0.5 feet and a standard deviation of 0.5 feet. What proportion of student are between 5.81 feet and 6.1 feet tall? $P(5.81 < x < 6.3) = ?$ (Given $P(z < 0.62) = 0.7324$ and $P(z < 1.6) = 0.9452$) (6.5)
- ii. Discuss sampling distribution and its types. (6)

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UNIT-II

Q4.

- i. To assess the significance of possible variation in performance in a certain test between the schools of a city, a common test was given to a number of students taken at random from 12 class of three schools concerned results are given below

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Make the analysis of the variance for the given data. (Tabulated value $F_{0.05}=3.89$) (6.5)

- ii. A coin was tossed 484 times and head turned up 265 times. Test the hypothesis that coin is unbiased. (6)

OR

Q5.

- i. From the following data find out whether there is any relationship between sex (M/F) and preference color

Color	M	F	Total
Pink	10	40	50
Black	70	30	100
Yellow	30	20	50
Total	110	90	200

Tabulated value $\chi^2_{0.05}=5.99$ (6.5)

- ii. A random sample of 50 items gives the mean 6.2 and variance 10.24 can it be regarded as drawn from a normal distribution with 5.4 at 5% level of significance. (6)

UNIT-III

Q6.

- i. Marks is a list that stores marks of a student in 10-unit test. Write a program to plot Line chart for the student's performance in these 10 tests. (6.5)

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- ii. Write a program to draw bar graph for the following data for the Medal tally of CWG-2023:- (6)

Gold	Silver	Bronze	Total
26	20	20	66

OR

Q7.

- i. Write a program to plot a horizontal bar chart from the height of some students. (6.5)
- ii. Explain how Seaborn's statistical visualization capabilities enhance data exploration and analysis in comparison to other Python visualization libraries? (6)

UNIT-IV

Q8.

- i. Write a python code for simple GUI calculator using Tk and perform various arithmetic operations. (6.5)
- ii. Write a Python program that designs a themed label widget with a different font, color, and background style using Tkinter. (6)

OR

- Q9. Create a GUI based form for admission purpose for your college. The form is connected to the database and various database operations (create, insert, delete, update) are performed (12.5)

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